

Standard Operating Procedure ***Drop Saw***

Drop Saw

A chop saw is a high-speed power tool designed to make straight, accurate crosscuts in metal or wood using a circular blade. Due to the high rotational velocity and exposed blade, it requires strict adherence to clamping and hand-placement protocols to prevent catastrophic injury.

Potential Hazards

- Be aware of the following potential hazards when working with Horizontal Band saws:
  - **Mechanical**: Point of operation (blade), rotating parts, and sparks/shards.
  - **Operational**: High heat at the cut zone, loud noise, and unstable stock.
  - **Energy Sources**: High-speed rotational kinetic energy and electrical power.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**Safety Glasses (ANSI Z87.1):** Required at all times to protect against metal shards and projectile debris.
Footwear	**Closed-Toe Shoes:** Steel-toed boots are recommended when handling heavy plate stock or large sheet material.
Hand Protection	**Heavy-Duty Work Gloves:** Advised when handling raw stock to prevent lacerations from sharp "burred" edges.
Clothing/Hair	Tie back long hair and secure loose clothing. Remove all jewelry, especially rings, which can be caught in material during the feed.



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Before Use

- **Location Check**: Ensure the work area is clear of flammable materials, tools, and debris.
- **Blade Inspection**: Verify the blade is the correct type for the material (Abrasive vs. Cold Saw) and is free of cracks or missing teeth.
- **Guard Integrity**: Ensure the retractable lower blade guard moves freely and returns to the "closed" position automatically.
- **Table Setup**: Check that the vise is functional and the table is free of small off-cuts that could become projectiles.

During Use

- **Work holding**: **All stock must be secured in the saw's vise.** Never attempt to hold the workpiece by hand during a cut.
- **The "4-Inch" Rule**: Keep hands at least 4 inches away from the blade's path at all times.
- **Start-Up**: Allow the saw to reach full RPM before engaging the material.
- **Cutting Pressure**: Do not force the blade through the stock. Let the RPM of the saw do the work to prevent blade "walking" or motor stall.
- **Spark Control**: Ensure the spark deflector is aimed away from other personnel or flammable bins.
- **Binding**: If the blade binds, release the trigger immediately and wait for the blade to stop before attempting to clear the stock.

After Use

- **Blade Stop**: Wait for the blade to come to a complete stop before lifting the head or reaching for the workpiece.



- **Power Down**: Release the lock-down pin (if equipped) to secure the saw head in the down position for storage.
- **Cleanup**: Use a brush (never hands) to clear metal chips and abrasive dust from the table.
- **Floor Safety**: Sweep up metal shards and debris around the machine base to prevent slipping or tracking fines into the shop.

Emergency Equipment and Procedures

- **E-Stop**: Familiarize yourself with the trigger-lock or main power switch location.
- **Reporting**: All blade failures, injuries, or near-misses must be reported to the Laboratory Project Supervisor immediately.